Volume 36

Sentinel AI Orchestration Framework (SAOF)

Status: Core Architecture

Priority: Critical

Vision

Artificial Intelligence should never become the source of truth.

It should become an intelligent assistant that operates on verified evidence and always explains its reasoning.

The Evidence Engine remains authoritative.

Design Philosophy

AI never owns data.

AI never owns trust.

AI never owns evidence.

AI never owns policies.

AI assists.

The platform decides.

The user approves.

AI Layer

```
User
```

Everything goes through the Orchestrator.

Why an AI Orchestrator?

Instead of integrating one LLM directly.

We support many.

Example

```
OpenAI
```

They become interchangeable.

AI Provider Interface

Every provider implements exactly the same interface.

Required capabilities:

- Chat
- Summarization
- Classification
- Translation
- Structured Output
- Function Calling
- Embeddings (optional)
- Image Understanding (optional)

This keeps the platform vendor-neutral.

AI Object

Every AI interaction becomes an object.

```
AI Request ID
```

Every request is traceable.

AI Modes

The user chooses how AI operates.

Assistant Mode

Provides explanations.

Never changes anything.

Advisor Mode

Produces recommendations.

Requires approval before actions.

Automation Mode

Can execute approved workflows.

Restricted to authorized environments and governed by RBAC, policies, and audit logging.

Explainable AI

Every answer includes:

Evidence Used

↓

Reasoning Summary

↓

Confidence

↓

Limitations

↓

Recommendation

The user can inspect the evidence.

AI Context Builder

Instead of sending an entire database.

The Context Builder selects only the relevant:

- Objects
- Evidence
- Policies
- Relationships
- Trust history
- Mission data

This improves both privacy and performance.

Privacy Engine

Organizations decide:

Never leave device

↓

Local AI only

or

Cloud AI Allowed

or

Hybrid

The platform adapts automatically.

Prompt Security

Every prompt passes through validation.

Checks include:

- Sensitive information.
- Prompt injection attempts.
- Unauthorized data requests.
- Policy violations.

This protects both the user and the organization.

AI Memory

We do **not** give AI unlimited memory.

Instead we maintain:

Conversation Context

Task Context

Mission Context

Evidence Context

Knowledge Context

Each has explicit boundaries and retention policies.

AI Workspace

Every workspace receives specialized guidance.

Example

SOC Workspace

↓

Incident explanation

↓

Timeline summary

↓

MITRE mapping

↓

Suggested investigations

Home Workspace

↓

Device health summary

↓

Simple explanations

↓

Next steps

The experience is adapted to the audience.

Multi-Language Support

Because we agreed Project Sentinel should support Norwegian, English, and additional languages.

AI helps:

- Translate interface content.
- Explain findings.
- Generate reports.
- Answer questions.

Security terminology remains consistent through a managed glossary.

NEW CORE ENGINE

AI Governance Engine

Responsible for:

- Provider approval.
- Model versions.
- Usage policies.
- Data residency.
- Audit logs.
- Human approval requirements.

Organizations retain full control.

NEW CORE ENGINE

AI Evaluation Engine

We should continuously measure AI quality.

Metrics include:

- Accuracy.
- Hallucination rate.
- Evidence usage.
- User feedback.
- Response consistency.

- Performance.

This allows us to improve prompts and provider selection without guessing.

NEW CORE ENGINE

AI Task Planner

Instead of one large prompt.

The planner decomposes complex requests.

Example:

User asks:

> Assess the security posture of our Finance department.

Planner creates tasks:

1. Inventory assets.
2. Collect evidence.
3. Evaluate policies.
4. Calculate trust.
5. Review findings.
6. Generate recommendations.
7. Produce executive summary.

Each task is evidence-backed.

NEW IDEA

AI Pair Investigation

Instead of replacing analysts.

The AI works beside them.

Analyst investigates.

↓

AI summarizes.

↓

Analyst verifies.

↓

AI prepares report.

↓

Analyst approves.

This keeps people in control.

NEW IDEA

AI Teaching Mode

For home users and junior analysts.

Instead of only answering:

> Enable BitLocker.

The platform explains:

- Why encryption matters.
- What changes after enabling it.
- How to verify success.
- How it affects Trust Score.

The platform becomes educational.

NEW IDEA

AI Architecture Advisor

A unique capability.

The AI understands the Project Sentinel architecture itself.

Developers can ask:

> Where should this new module be added?
>
> Which engine should own this feature?
>
> Which ADR applies?

This supports long-term architectural consistency.

PS-ADR-0031

AI Assists, Evidence Decides

Status: Accepted

Decision

Artificial Intelligence shall augment user understanding and operational efficiency but shall never replace evidence, governance, or human accountability for security decisions.

Rationale

This preserves trust, transparency, auditability, and regulatory compliance while allowing AI capabilities to evolve independently.

NEW LONG-TERM VISION

Sentinel Intelligence Mesh

I think we can already see another major architectural layer emerging.

Rather than one AI model, Project Sentinel can orchestrate a **mesh of specialized intelligence services**:

- Natural language understanding.
- Document analysis.
- Image analysis.
- Code analysis.
- Malware triage.
- Threat intelligence enrichment.
- Translation.
- Report generation.

Each service is replaceable and communicates through the same Object and Event architecture.

A proposal that could become one of our strongest competitive advantages

I believe Project Sentinel should include an **AI Transparency Panel** on every screen where AI contributes.

It would answer:

- **Which evidence did the AI use?**
- **Which model generated the response?**
- **When was it generated?**
- **How confident is the response?**
- **Can I verify every conclusion?**

Many platforms simply display AI-generated text. We can do better by making AI contributions fully inspectable and grounded in evidence.

Looking Ahead

The next volume should be **Volume 37 – Sentinel Distributed Scan Engine (SDSE)**.

This will define how Project Sentinel securely coordinates scanning across:

- Home computers.
- Enterprise fleets.
- Mobile devices.
- Remote offices.
- Cloud environments.
- Edge devices.

It will allow organizations to assess thousands—or even hundreds of thousands—of authorized assets in parallel while using the same Mission, Object, Evidence, and Workflow architecture we've established. This is also where we will define how Project Sentinel scales from protecting a single family laptop to securing the infrastructure of banks, governments, healthcare providers, and global enterprises without changing its core architecture.

=====[USER]=====

Good and continue

=====[ASSISTANT]=====

Excellent. We have now designed the brain of Project Sentinel.

Now we design the muscles.

This volume defines **how Project Sentinel performs security assessments at any scale**—from one home computer to hundreds of thousands of enterprise assets.

One important clarification before we begin:

Our platform will **only** perform active testing against systems that the user owns or is explicitly authorized to assess (for example, assets covered by an organization's security program or an authorized bug bounty scope). That keeps the platform aligned with professional and legal security practice.

PROJECT SENTINEL BLUEPRINT

Volume 37